Encoding a decoded circle code returns the original integer

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 591-594 and its proof environment opens at line 596 after a blank line; the pairing is by adjacency and the retained proof range 596-623 is the whole environment.

Editorial note. The retained counter values (chapter 8, section 1, theorem offset 7) make the counted lemma print as Lemma 8.1.8, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel lem:s1-encode-decode = 8.1.8).

Editorial note. Two labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: thm:sign-induction (book Theorem 6.10.12, defined at line 1399 of the hits chapter pinned file and counted by its own card of this lane) and lem:transport-s1-code (book Lemma 8.1.2, defined at line 447 of this chapter pinned file).

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the encode-decode identity for the circle code at an arbitrary point (author label lem:s1-encode-decode, book Lemma 8.1.8).

Allowed standard prerequisites (editorial).

- The circle code family. The code family assigns to each point of the circle a type, with the integers at the base point, and decode turns an element of the code into a path from the point to the base point.. They are the two maps composed in the statement.
- Induction on the integers. A type family over the integers with a point at zero and steps in both directions has a section.. It is the induction that runs the main computation.
- Transport along the generating loop adds one. Transporting in the code family along the generating loop adds one to the integer, as computed for the circle code.. It is the step that turns the transport chain into arithmetic.

Source reference. Original theorem 6.10.12: sources/hott-hits/pinned-original-source.tex, source lines 1399–1399. Book Theorem 6.10.12, the two-sided induction principle for the integers, invoked at source line 606 for the induction that

carries the computation. Navigation pointer only: it is counted by its own card of this lane and the retained proof carries out the induction it licenses.

Source reference. Original lemma 8.1.2: sources/hott-homotopy/pinned-original-source.tex, source lines 447–447. Book Lemma 8.1.2, the computation of transport in the circle code along the generating loop, used at the fourth line of the retained computation. Navigation pointer only.

Lemma 8.1.8. *For all* $x : \mathbb{S}^1$ *and* $c : \mathsf{code}(x)$*, we have* $\mathsf{encode}_x(\mathsf{decode}_x(c)) = c$.

Proof. The proof is by circle induction. It suffices to show the case for base, because the case for loop is a path between paths in $\mathbb{Z}$, which is immediate because $\mathbb{Z}$ is a set.

Thus, it suffices to show, for all $n : \mathbb{Z}$, that

$$\mathsf{encode}'(\mathsf{loop}^n) = n.$$

The proof is by induction, using original theorem 6.10.12.

- In the case for 0, the result is true by definition.
- In the case for $n + 1$,

$$\begin{aligned}
\mathsf{encode}'(\mathsf{loop}^{n+1}) &= \mathsf{encode}'(\mathsf{loop}^n \cdot \mathsf{loop}) && \text{(by definition of } \mathsf{loop}^-) \\
&= \mathsf{transport}^{\mathsf{code}}((\mathsf{loop}^n \cdot \mathsf{loop}), 0) \\
& && \text{(by definition of } \mathsf{encode}) \\
&= \mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, (\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^n, 0))) \\
& && \text{(by functoriality)} \\
&= (\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^n, 0)) + 1 \\
& && \text{(by original lemma 8.1.2)} \\
&= n + 1. && \text{(by the inductive hypothesis)}
\end{aligned}$$

- The case for negatives is analogous. □